



TARGET-90

JKBOSE CLASS-11TH

Anatomy of Flowering Plants

One Shot

BIOLOGY

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Topics To Be Covered

1 One Shot

2 ...

Introduction

PLANT ANATOMY: Study of internal structures of plant is called plant anatomy.

- Internal structure of plants shows the adaptations towards diverse environments.
- In Angiosperms, Monocots and Dicots are anatomically different.

Overview of Plant Tissue

- Term tissue was coined by "Grew".
- Study of tissues is called Histology.
- "A group of cells having common origin and usually performs common function
is called as tissue."

NOTE:

- Structure of cells of a tissue may be same or different.

Ex. Same → Nervous tissue, *parenchyma*, *collenchyma*, *Sclerenchyma*.

Different → Xylem, Phloem etc.

Plant Tissue

2 types

1) Meristematic Tissue (Dividing)

Ex: SAM, RAM, Vascular Cambium
of Root & Stem,
Intercalary meristem
Axillary meristem

2) Permanent tissue (Non-dividing)

→ Mature tissue.

2 types

a) Simple permanent tissue

→ All cells are
similar in str.

Ex: Parenchyma, Collenchyma,
Sclerenchyma

b) Complex permanent tissue

→ Composed of more
than one type of cells.

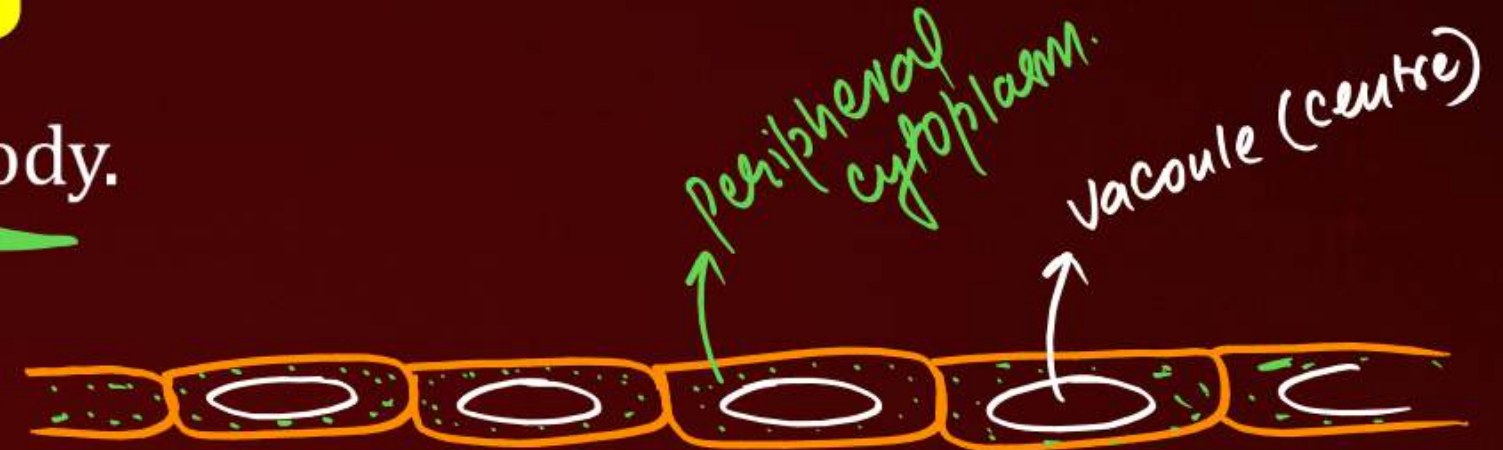
Ex: Xylem, Phloem.

Tissue System

- In plants, tissues combine to form tissue system.
- Structure and function of a tissue depends on its location.
- On the basis of structure and location, 03 types of tissue systems are present.
 1. Epidermal tissue system. ✓
 2. Ground tissue system / Fundamental tissue system. ✓
 3. Vascular tissue system / Conducting tissue system. ✓

1. Epidermal Tissue System

- It forms outermost covering of plant body.
- It comprises:
 - a) Epidermal cells. ✓
 - b) Epidermal str. ✓
 - c) Epidermal appendages. ✓



a) Epidermal cells:

- It forms epidermis which is outermost protective covering of plant body.
- Elongated & compactly arranged cells (No intercellular spaces).
- Epidermis — usually single & continuous layer.
- Epidermal cells → Parenchymatous having large vacuole, peripheral cytoplasm

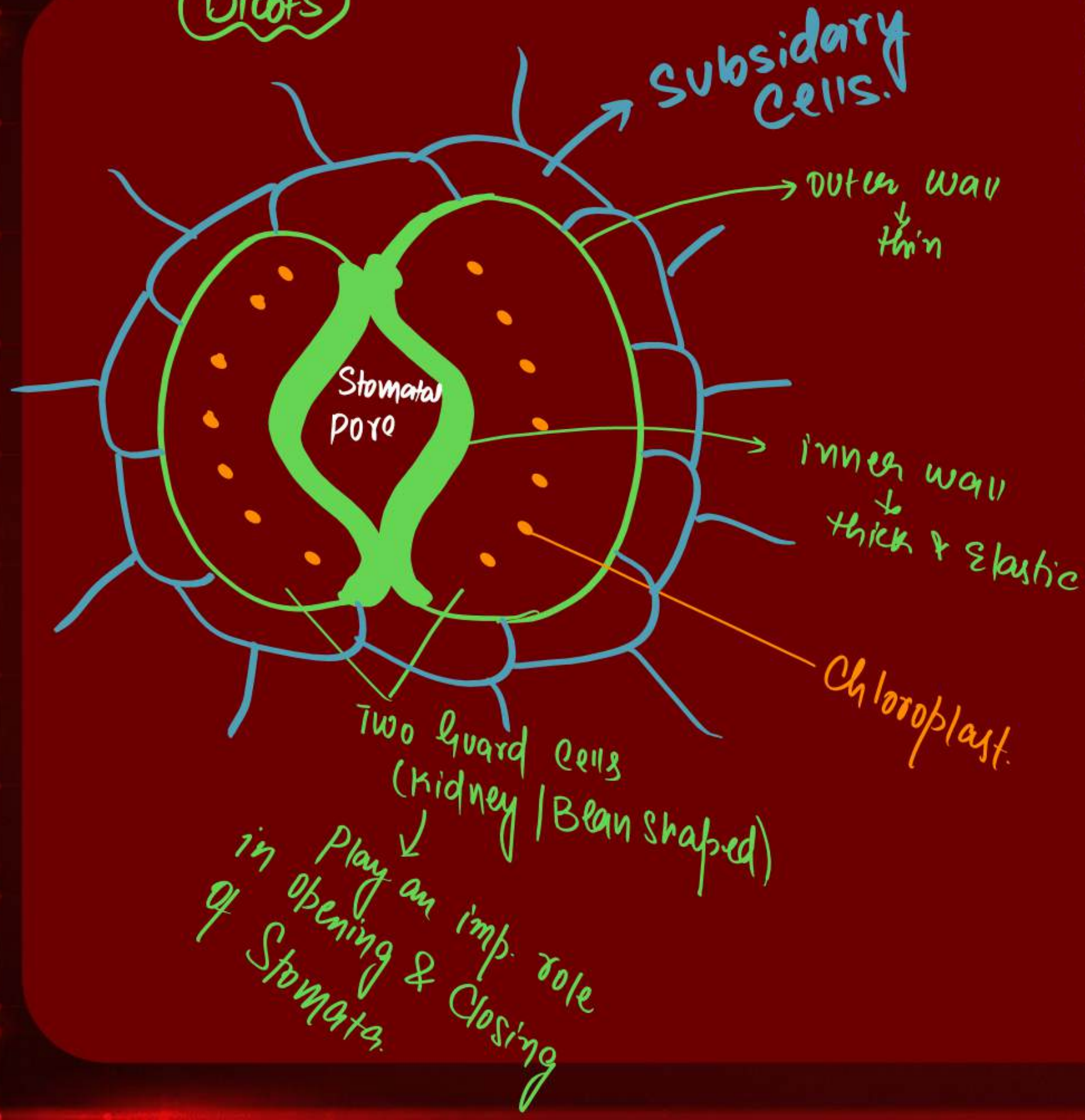
Note: Chloroplast is usually absent in Epidermal cells.

1. Epidermal Tissue System

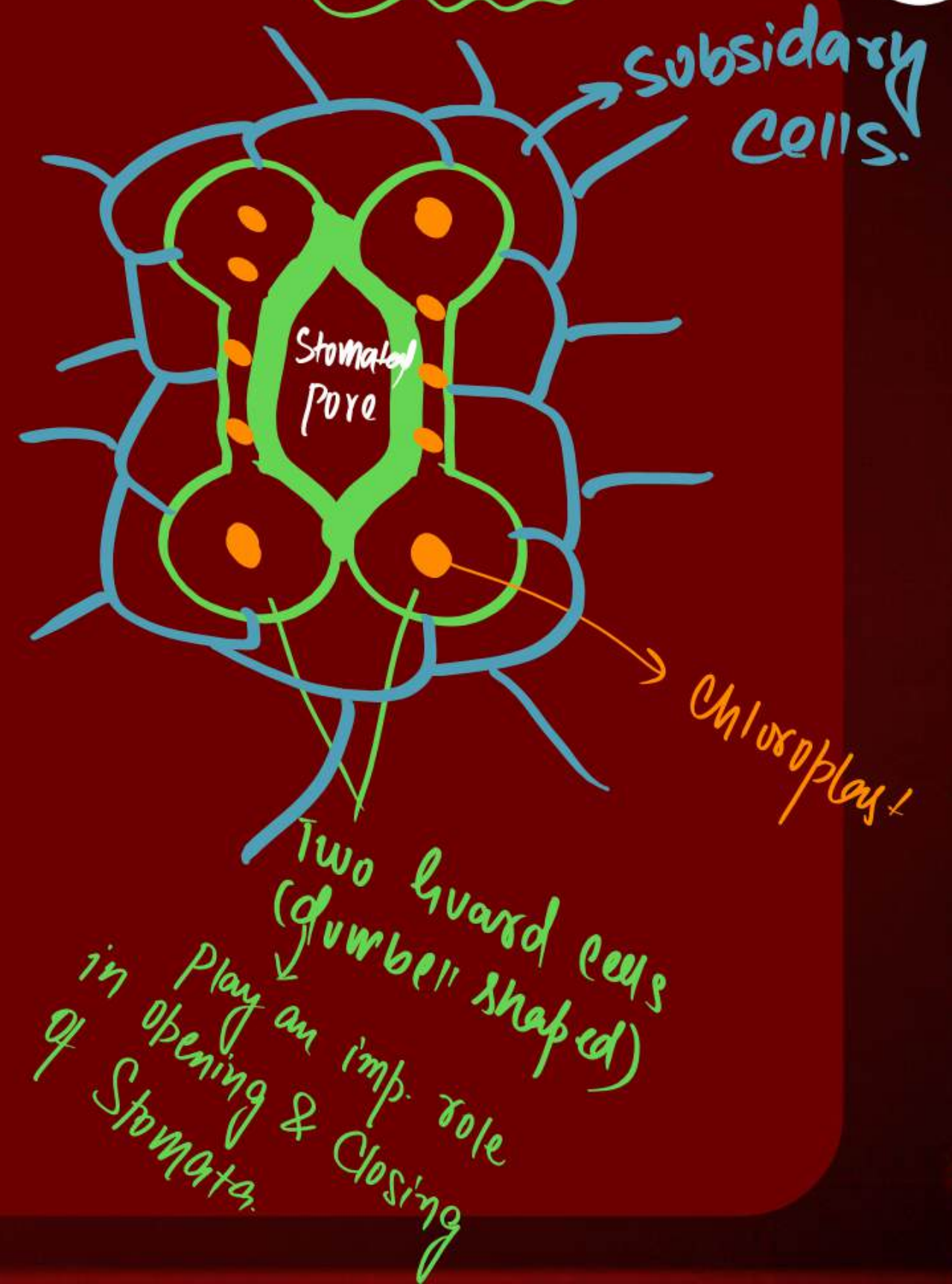
Epidermis is externally covered by a waxy thick layer called as **CUTICLE**

- prevents H_2O loss.
- present in herbaceous stem & leaves.
- absent in Roots & woody Stems.

Dicots



Monocots



b) Epidermal Structure

- It consists of Stomatal apparatus. (present abundantly in the epidermis of leaves).
- Stomatal apparatus consists of:

- a) Two guard cells
- b) Subsidiary cells
- c) Stomatal pore

a) Two guard cells:

- Only epidermal cells which possess chloroplast
- Regulates opening & closing of stomata

b) Epidermal Structure

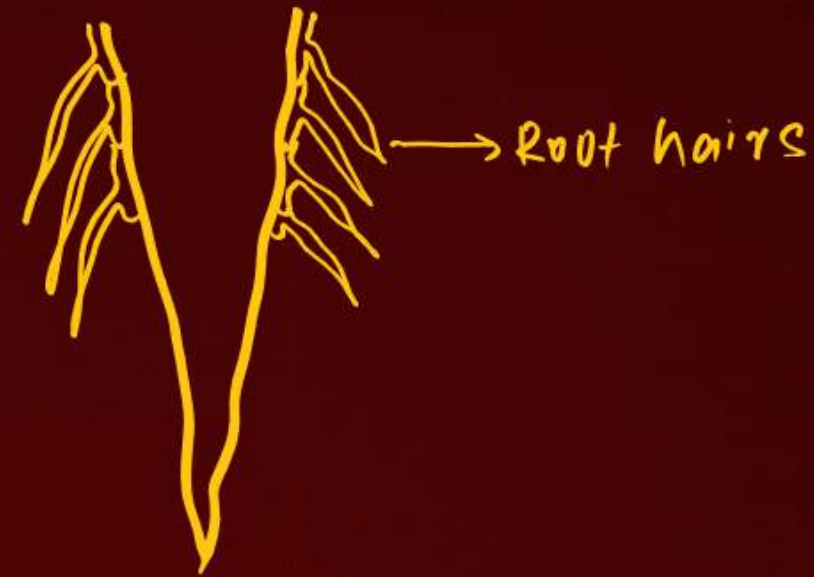


b) Subsidiary cells:

- Specialized epidermal cells around guard cells.
- Parenchymatous & lacks chloroplast.

c) Epidermal Appendages

- It includes
 - Root hairs
 - Stem hairs or Trichomes.



Root hairs:

- Unicellular elongations of epidermal cells of root (of maturation zone).
- Increases surface area for H₂O and mineral absorption from soil.

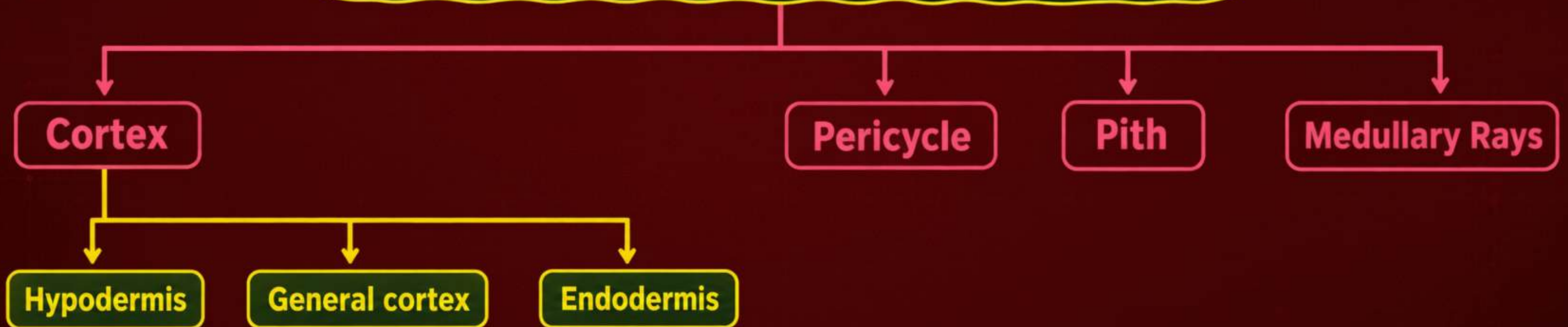
Stem hairs or Trichomes:

- Usually multicellular. (or may be unicellular)
- Branched / unbranched.
- Soft / stiff. helps in secretion & prevents H₂O loss due to transpiration.

2. Ground Tissue System

- It consists of all tissues except — Epidermis and Vascular Bundle.
- It consists of simple tissues like — Parenchyma, Collenchyma, Sclerenchyma.

Ground tissue system is differentiated into:



Note: In leaves — ground tissue is not differentiated which is called as Mesophyll.



3. VASCULAR TISSUE SYSTEM

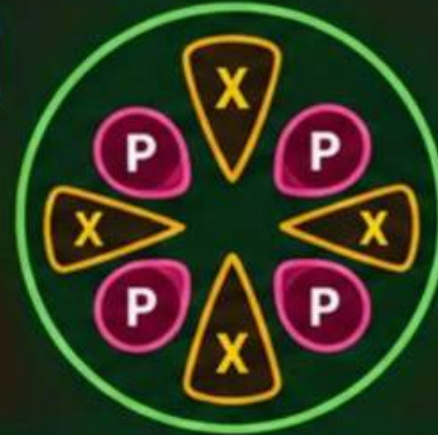
→ Consists of **xylem & phloem**.

2 types

Radial vascular bundle

Ex: Root

- **Xylem & phloem** are arranged on different radii in alternating manner.



Conjoint vascular bundle

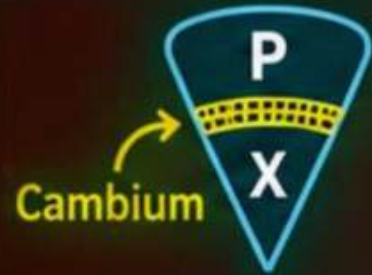
- **Xylem & phloem** are present together on the same radius of vascular bundle.
- **Ex:** Stem & leaves



Open conjoint collateral vascular bundle

(Open V.B)

- Cambium is present between xylem and phloem.
- **Ex:** Dicot stem



Closed conjoint collateral vascular bundle

(Closed V.B)

- Cambium is absent between xylem and phloem.
- **Ex:** Monocot stem, Leaf



Conjoint collateral vascular bundle

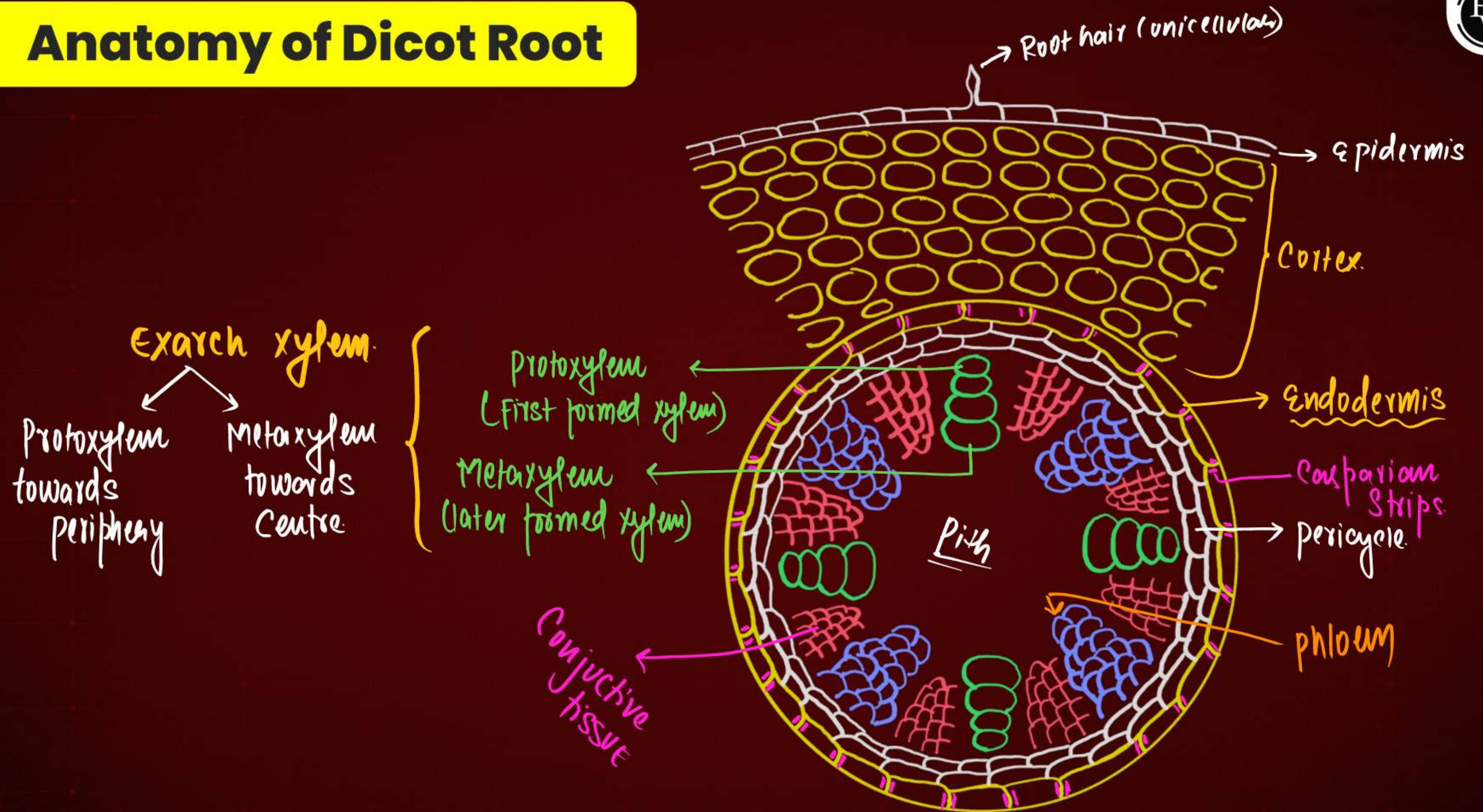
- Most common.
- Xylem towards inner side (centre).
- Phloem towards outer side (periphery).



2 types

“Vascular tissue forms the transport system of plants.”

Anatomy of Dicot Root



Anatomy of Dicot Root

Epidermis: Outermost layer, single layered, possess unicellular root hairs.

Cortex: Inner to epidermis, multilayered, parenchymatous (thin walled), intercellular spaces present.

Endodermis: Innermost layer of cortex, single layered, barrel shaped, no intercellular spaces, radial and tangential walls possess Casparian strips (water impermeable waxy material made up of suberin).

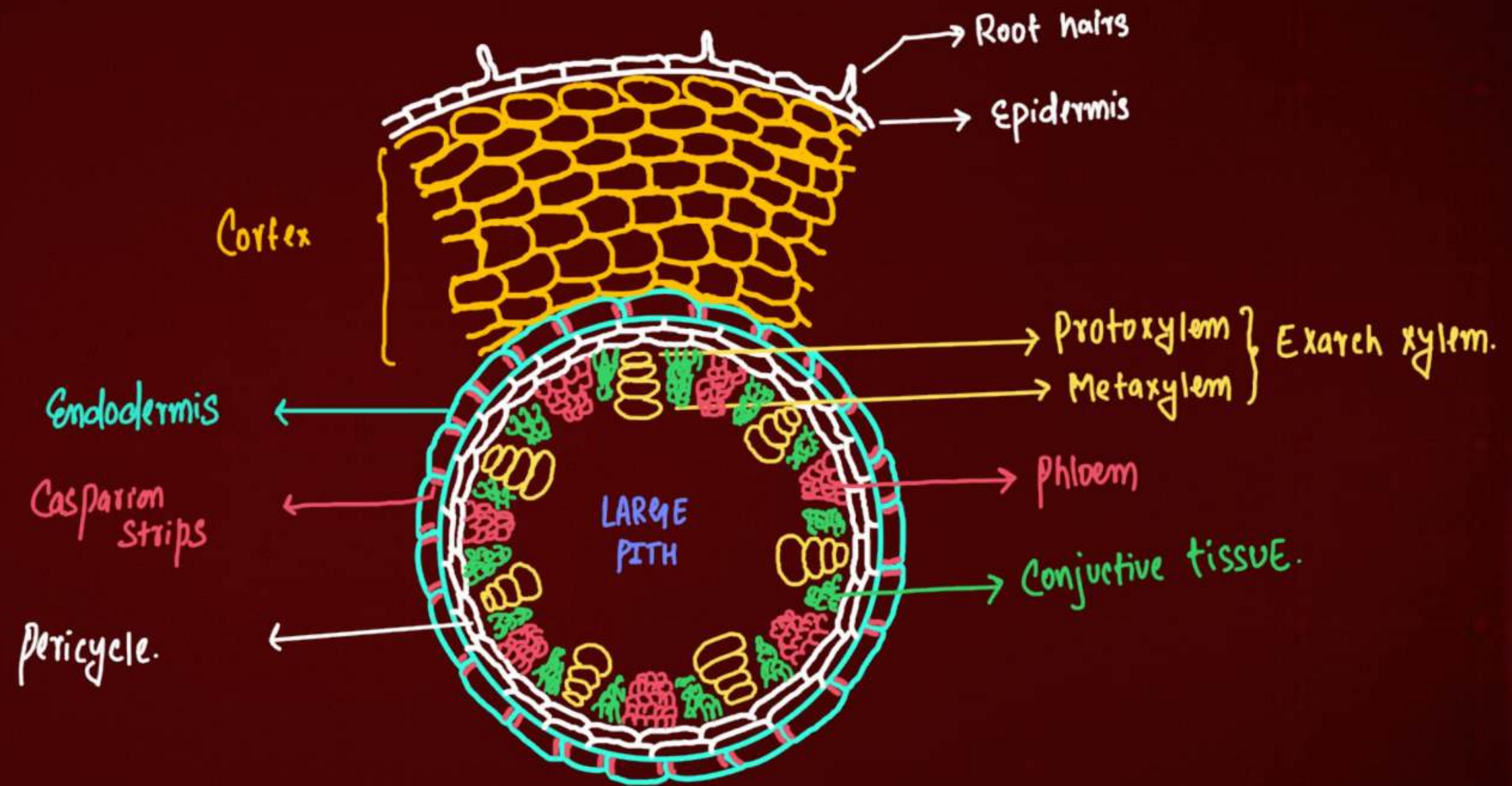
Pericycle: Inner to endodermis, few layered, parenchymatous (thick walled), forms lateral roots and cambium during secondary growth.

Vascular Bundle: Radial, 2-4 xylem and phloem patches, xylem is exarch

Conjunctive Tissue: Parenchymatous, present between xylem and phloem.

Pith: small, inconspicuous (small)

Anatomy of Monocot Root

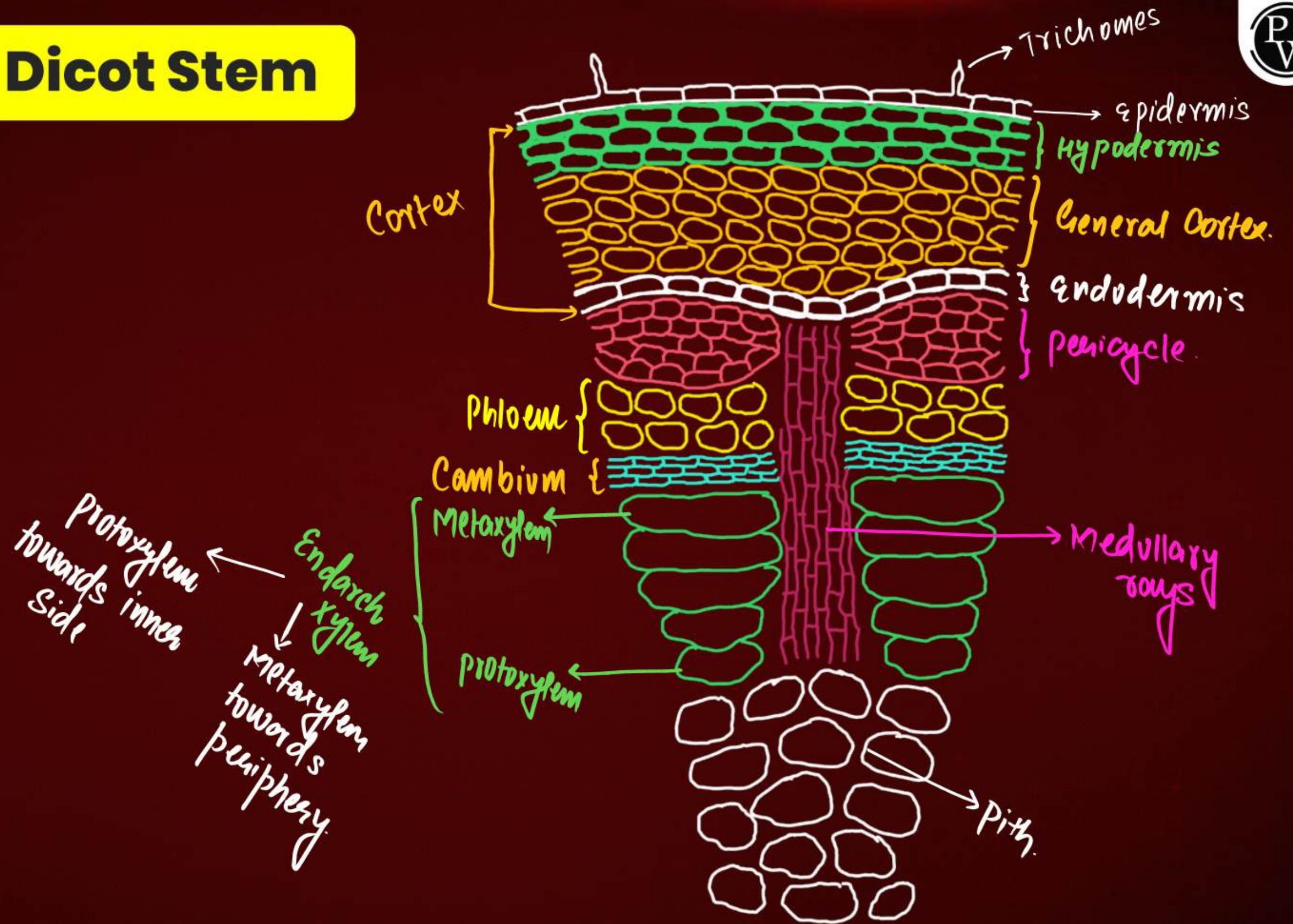


Anatomy of dicot root and monocot root is similar in many aspects except:

- 1) **VASCULAR BUNDLE:** radial, xylem and phloem patches are more than 6 (polyarch condition).
- 2) **PITH:** Large and well developed.

No secondary growth in monocot root.

Anatomy of Dicot Stem



Anatomy of Dicot Stem

EPIDERMIS: Outer most single layered, protective in nature, bear multicellular trichomes and few stomata.

CORTEX: Present between epidermis and pericycle, multilayered, divided into 3 sub-zones:

- a) **Hypodermis:** present below epidermis, few layered, collenchymatous in nature, intercellular spaces absent, provide mechanical support to young stem.
- b) **General cortex:** present below hypodermis, rounded thin walled and parenchymatous, multilayered, conspicuous intercellular spaces present.
- c) **Endodermis:** innermost layer of cortex, single layered, possess starch grains i.e., why it is also called as starch sheath.

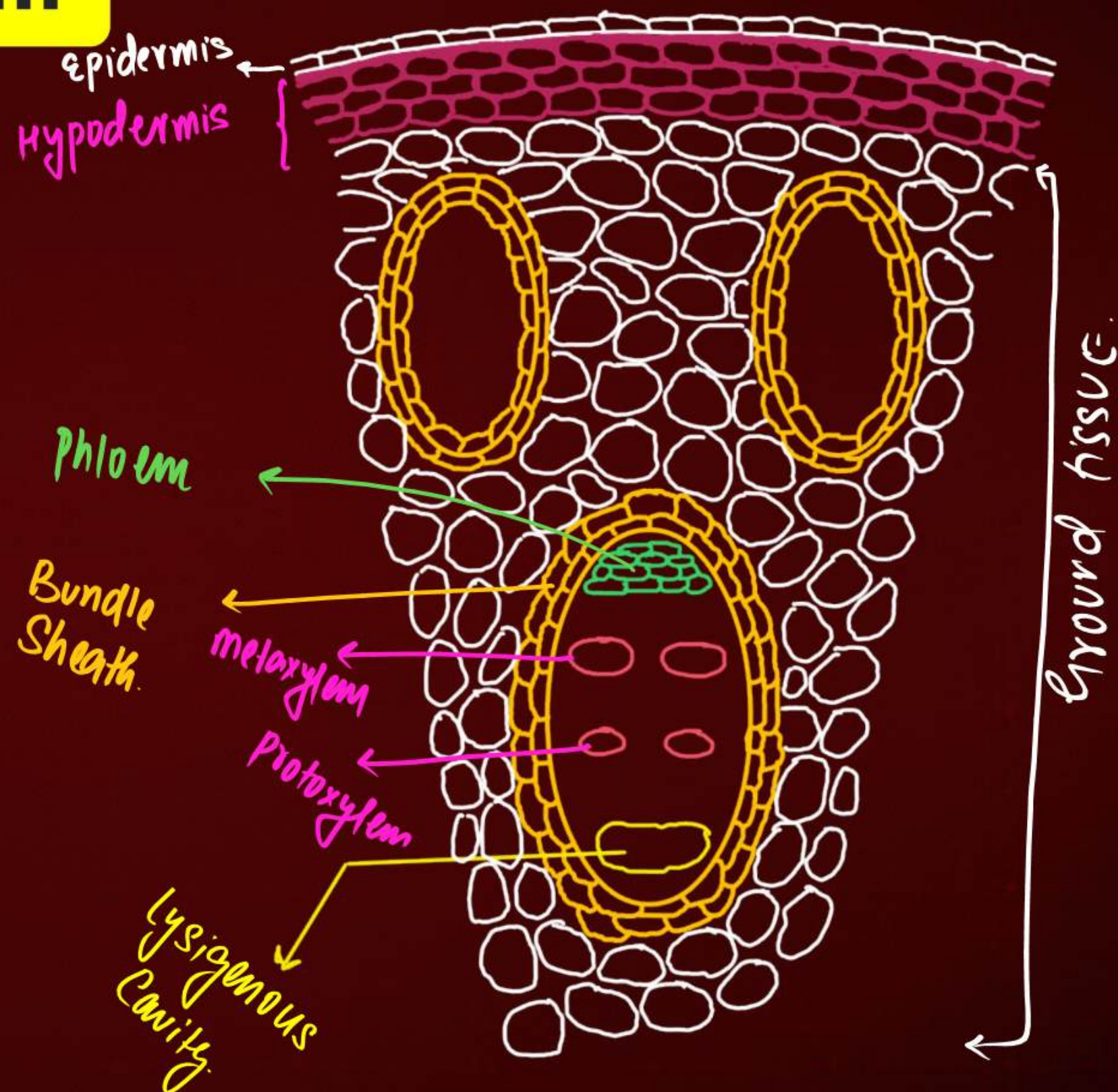
Anatomy of Dicot Stem

VASCULAR BUNDLE: large in number, arranged in ring (characteristic feature of dicot stem), vascular bundle is conjoint and open, xylem is endarch.

MEDULLARY RAYS: parenchymatous, few layered, present between vascular bundles.

PITH: central portion of stem, contains large number of rounded and parenchymatous cells, large intercellular spaces are present.

Anatomy of Monocot Stem



Anatomy of Monocot Stem

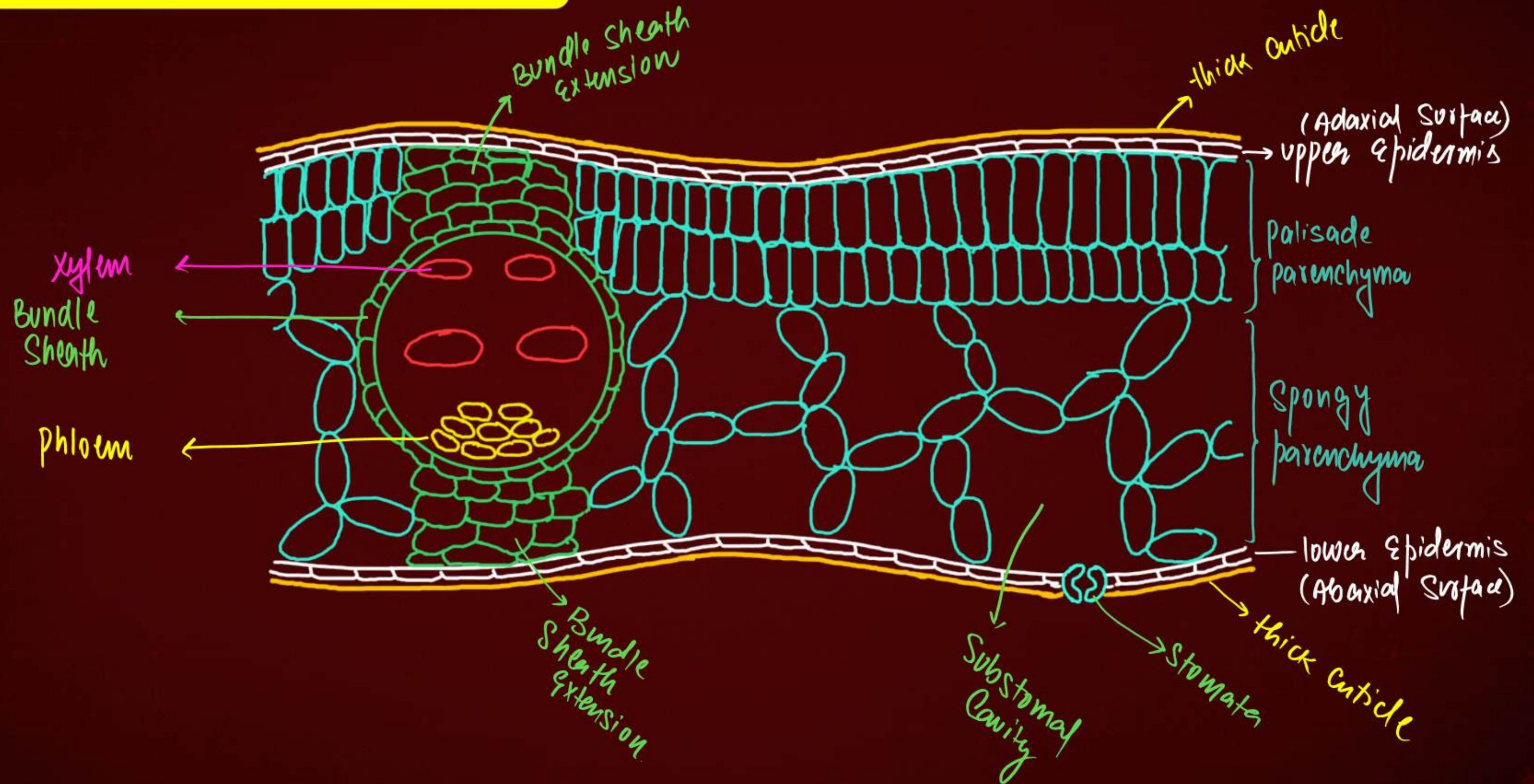
EPIDERMIS: outermost single layered, possess thin waxy layer (cuticle), trichomes are absent.

HYPODERMIS: present inner to epidermis, sclerenchymatous in nature.

GROUND TISSUE: present inner to hypodermis, parenchymatous and large cells.

VASDCULAR BUNDLE: present scattered inside ground tissue, abundant in number, small vascular bundles towards periphery and large towards centre, each vascular bundle is surrounded by sclerenchymatous sheath, vascular bundles are conjoint and closed, xylem is endarch, vascular bundles possess lysigenous cavities which stores water, gasses and essential oils.

Anatomy of Dicot Leaf



Anatomy of Dicot Leaf

EPIDERMIS: It covers both upper and lower surfaces, upper epidermis possess less or no stomata, lower surface possess more stomata, Epidermis possess thick cuticle.

MESOPHYLL: Tissue present between upper and lower epidermis, possesses chloroplast, parenchymatous, performs photosynthesis.

- It has two types of cells:

a) Palisade parenchyma- towards adaxial surface, elongated, arranged vertically, parallel to each other.

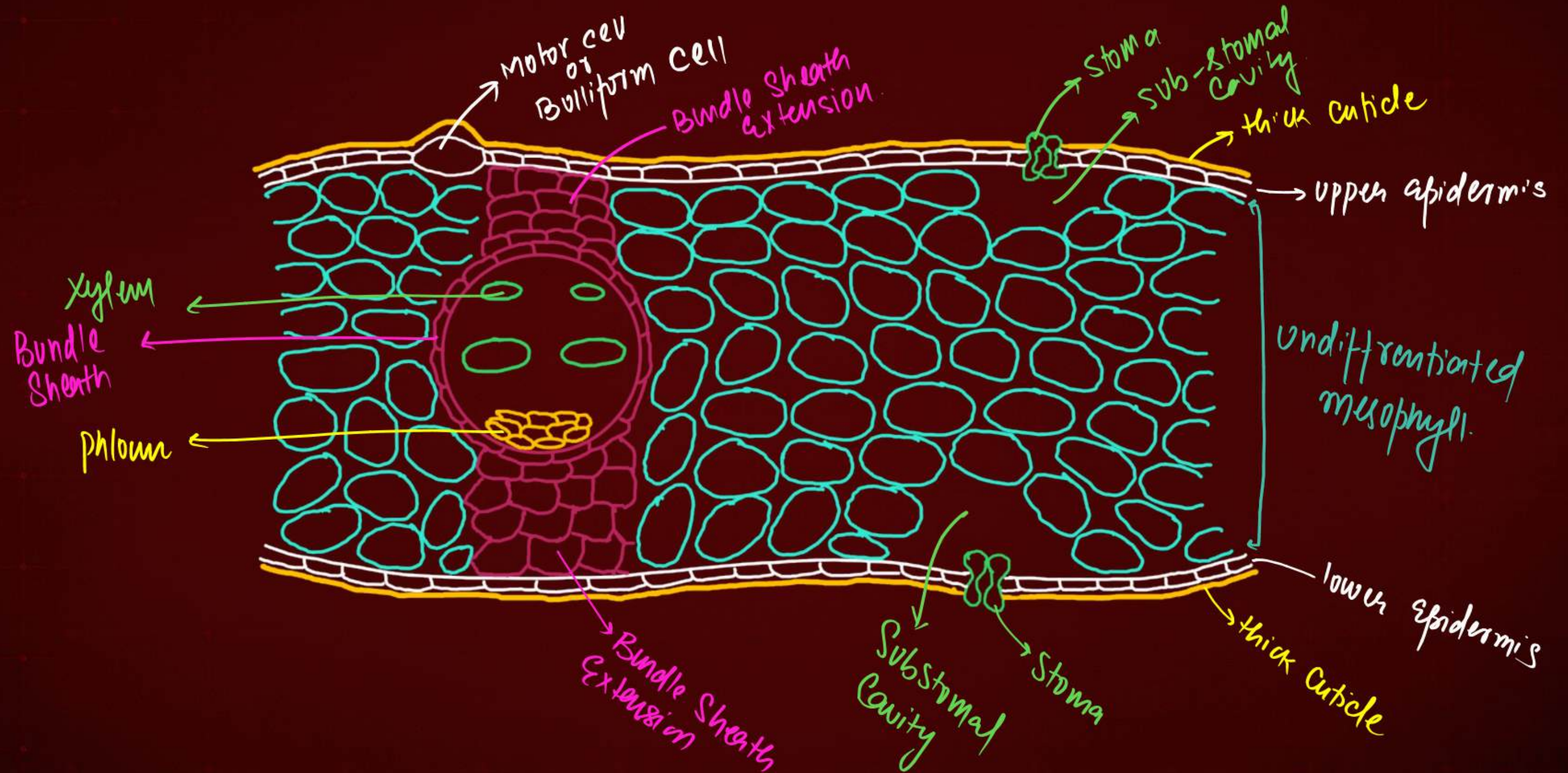
b) Spongy parenchyma- towards adaxial surface, oval/round, loosely arranged, possess air spaces and air cavities.

VASCULAR BUNDLE: seen in veins and midrib.

Anatomy of Dicot Leaf

- Size of VB depends on size of veins. In reticulate venation veins vary in thickness. Vascular bundles are surrounded by layer of thick-walled cells called as bundle sheath cells.
- Vascular bundles – conjoint, collateral, closed.
- Xylem towards upper surface and phloem towards lower surface.

Anatomy of Monocot Leaf



Anatomy of Monocot Leaf

EPIDERMIS: It covers both upper and lower surface.

- Almost equal number of stomata are present on both surfaces (isostomatic).
- Cuticle covers both the surfaces.
- On upper epidermis, bulliform cells / motor cells are present. These cells are large, empty, colourless.
- Due to water absorption, these cells become turgid & makes the leaf exposed.
- Due to water loss, these cells become flaccid & makes the leaf to curl inwards to minimize water loss.

Anatomy of Monocot Leaf

MESOPHYLL: parenchymatous, present between two epidermal layers, not differentiated into palisade and spongy, possess chlorophyll, performs photosynthesis.

VASCULAR BUNDLE:

- Thickness of vascular bundles are similar due to parallel venation (except in midrib).
- Surrounded by bundle sheath.
- Vascular bundle conjoint and closed.

Most IMPORTANT Questions for JKBOSE 2026

- Q1. Epidermal tissue system? ✓
- Q2. Vascular tissue system? ✓
- Q3. Discuss (along with diagrams):
 - a) Monocot & Dicot root. ✓
 - b) Monocot & Dicot stem. ✓

HOMework



Revision



Thank
You